GHOSTMESH

LLM Chain-Link Protocol // QNVM Sovereign Node

Technical Reference Document

Functions · Formulas · Equations · Features

CLCP-QNVM REV 1.0 | DNDAF/DoDAF/NAFv4 Compliant

DOCUMENT CLASSIFICATION: INTERNAL RESEARCH

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1. Entropy System

The entropy system is the foundational measurement layer of the GhostMesh platform. It governs stability, creativity, and the triggering conditions for advanced operational modes. All entropy values are normalised to the range [0.0, 1.0].

1.1 Shannon Log Entropy (S_log)

The primary metric harvested from Git commit logs. Computed via Shannon entropy over the probability distribution of commit event types.

```
S_log = -SUM [ P(event_i) * log2( P(event_i) ) ]
```

where P(event_i) is the probability of event type i across the harvested commit window. Event types include: commit type (feat/fix/refactor), file change count bucket, author identity, and hour-of-day band.

1.2 Entropy Bounds Constraint

The system enforces hard bounds on S_log to maintain creative coherence. Values outside the permitted range trigger agent rebalancing.

```
0.10 <= S_log <= 0.30  (nominal operating window)
```

Lower sub-bound 0.15 and upper sub-bound 0.25 define the crystalline zone. Breach of outer bounds halts the current RecursiveForge cycle immediately.

1.3 Entropy Increase Rate

Per-generation entropy increase applied to each entity in the QNVM substrate:

```
entropy(t+1) = entropy(t) + ENTROPY_INCREASE_S5 + entropy_mod
```

ENTROPY_INCREASE_S5 = 0.0004 (default). entropy_mod is applied by Co-Evolution Council proposals and Reality Editor edits. Minimum floor: 0.15.

1.4 Predictive Entropy Equation

Forecasts the log entropy of the next harvest cycle based on the current excitement score, enabling proactive agent rebalancing.

```
S_log(t+1) = S_log(t) + alpha * S_exc(t)
```

alpha is the Creative Inspiration coefficient, learned by a meta-learning sub-agent from long-term project goal alignment. Typical alpha range: 0.01 – 0.08.

1.5 Dashboard Entropy Mirror

The live dashboard metric directly feeds into the QNVM entity drift values, creating a real-time bidirectional link between the PHP layer and the simulation substrate.

```
drift_seed = (S_rec - 0.15) * 100
```

This scaled value is injected as the initial drift perturbation for all entities in the current generation.

---

2. Bloom Phase System (S_bloom)

The Bloom Phase is the primary thermodynamic pressure indicator of the system. It measures the rate of information densification relative to population coherence, and determines when phase transitions (including auto-refactoring and Omega rebirth) are triggered.

2.1 Core Bloom Equation

```
S_bloom = 369 * (dI/dt) * C
```

dI/dt = rate of information increase per generation cycle. C = mean population coherence (0.0–1.0). The constant 369 encodes the 3-6-9 harmonic resonance fundamental frequency.

2.2 Military-Extended Bloom

The military-grade variant incorporates DoDAF compliance weighting and the 420M-qubit Hilbert space multiplier via tensor-network compression scaling.

```
S_mil = 369 * (dI/dt) * C * K_DoDAF
```

K_DoDAF is the fractional completeness of auto-generated DoDAF/NAFv4 viewpoints (0.0–1.0). Full compliance yields K_DoDAF = 1.0, maximising the bloom signal.

2.3 QNVM-Scaled Military Bloom

The full quantum-scaled variant applies a tensor network compression multiplier derived from the 420M effective qubit count:

```
S_mil-qnvm = 369 * (dI/dt) * C * K_DoDAF * log2(1 + N_ent * D_sym)
```

N_ent = QNVM population count. D_sym = mean symbolic density across entities. The log2 term compresses the exponential Hilbert space expansion into an operationally tractable signal.

2.4 Bloom-Triggered Auto-Refactor

When S_bloom exceeds the refactoring threshold, a specialised code-refactoring agent is dispatched autonomously via the CI/CD pipeline hook:

```
if (S_bloom > BLOOM_THRESHOLD && dI/dt > D_THRESH)
{ trigger_refactor_agent(); }
```

BLOOM_THRESHOLD default: 500. D_THRESH default: 1.5 units/cycle. The refactor agent is drawn from the active LLM Swarm, not a static script.

2.5 Bloom Phase → TrickleCharge Coupling

```
DW_budget(gen) = 5.0 + 0.1 * (gen - PHASE2_END)
```

In Phase 3 (gen > 30), the dark wisdom trickle-charge budget scales linearly. S_bloom values above 500 trigger an immediate one-full-segment DW release from the DarkWisdomExtractor payload.

---

3. Excitement Score (S_exc)

The excitement score quantifies the novelty and surprise value of outputs produced by the Rivalry Arena and Recursive Forge. It drives the creative injection cycle and correlates with subsequent entropy spikes.

3.1 Lexical Diversity Model

```
S_exc = w_novelty * N(text) + w_diversity * L(text) + w_surprise *
X(text)
```

N(text) = novelty score vs. blueprint history. L(text) = type-token ratio (lexical diversity). X(text) = semantic distance from centroid of current generation outputs. Weights: w_novelty=0.5, w_diversity=0.3, w_surprise=0.2.

3.2 Dual-Fitness Arena Equation

The Rivalry Arena uses a context-sensitive fitness function that shifts between creative and practical modes based on the current operational state:

```
F = isFunMode ? (w_U*U + w_R*R) : (w_N*N + w_E*E)
```

U = Utility score, R = Robustness score, N = Novelty score, E = Excitement score. Fun Mode (3-6-9 resonance active): w_U=0.55, w_R=0.45. Normal Mode: w_N=0.6, w_E=0.4.

3.3 3-6-9 Resonance Scheduler

Arena duels that occur on generation numbers divisible by 3, 6, or 9 receive a resonance boost that activates Fun Mode and increases the probability of Omega-class blueprint generation:

```
if (gen % 3 == 0 || gen % 6 == 0 || gen % 9 == 0)
arena.setResonanceMode(true);
```

Resonance duels contribute a +0.15 bonus to the winner's excitement score and commit their output with a DoDAF XML wrapper automatically.

3.4 Excitement-Entropy Resonance Correlation

```
S_log(t+1) = S_log(t) + alpha * S_exc(t)    [from Section 1.4]
```

Empirically observed: peak S_exc in the Arena predicts a S_log spike of +0.03 to +0.07 in the immediately following harvest cycle. High excitement artifacts are intentionally re-seeded into the forge pool to sustain elevated entropy.

---

4. Sophia Score & Hardlock System

The Sophia score is the primary fitness and coherence quality metric for every QNVM entity. It is anchored to the inverse of the golden ratio (INV_PHI) as the attractor state, representing the optimal balance between intelligence, coherence, and entropy.

4.1 Golden Ratio Constants

```
PHI       = (1 + sqrt(5)) / 2  =  1.6180339...
```

The golden ratio — used as the structural attractor baseline.

```
INV_PHI   = 1 / PHI            =  0.6180339...
```

The inverse golden ratio — the Sophia hardlock target. All viable entities have coherence pulled toward this value.

```
PHI_SQ    = PHI^2              =  2.6180339...
```

Used in advanced polytope geometry calculations within the D4 tensor.

4.2 Core Sophia Score Formula

```
sophia = max(0, min(1, (1 - |coherence - INV_PHI| * 2) * (IQ / 100) *
(1 - entropy)))
```

Three-factor product: (a) proximity of coherence to INV_PHI, (b) normalised intelligence, (c) inverse entropy. A perfect Sophia score of 1.0 requires coherence = 0.618, IQ = 100, entropy = 0.0.

4.3 Sophia Hardlock Initialisation

```
coherence_init = 0.60 * INV_PHI + 0.40 * archetype_coherence_base +
U(-0.03, 0.03)
```

New entities are blended 60% toward the hardlock attractor and 40% toward their archetype profile, with uniform noise. This ensures population-level coherence converges faster.

4.4 Sophia Fitness Component

```
sophia_fitness = max(0, 1 - |coherence - INV_PHI| * 2) * 0.15
```

Additive contribution to survival_prob(). Sovereign entities may receive additional +0.20 survival boost on top of this.

4.5 Phi-Pull (Coherence Restore)

Every generation, each entity's coherence is pulled toward INV_PHI by a proportional restoration force:

```
coherence(t+1) = coherence(t) + (INV_PHI - coherence(t)) * 0.10 +
coherence_mod
```

The 0.10 pull rate ensures convergence within approximately 20 generations from any starting coherence. coherence_mod is supplied by the DarkWisdomFurnace and Co-Evolution Council.

4.6 Sophia Point Detection

The Sophia Point is the system-level attainment condition: a measurable database flag set when the entire active population meets crystalline compliance criteria simultaneously.

```
isSophiaPoint() = (S_rec < 0.15) AND (K_DoDAF >= 1.0) AND (avg_sophia >=
INV_PHI - 0.05)
```

---

5. QNVM Entity Dynamics

The Quantum Neural Virtual Machine simulates a population of cognitive entities (Entities) across discrete generations. Each entity maintains a rich state vector governing its cognitive, ethical, and physical properties.

5.1 Survival Probability

```
P_surv = coherence * (1 - entropy) + w_IQ * (IQ/100) + sophia_fitness
+ diversity_bonus + sovereign_bonus + metacog_bonus
```

w_IQ = 0.30 (configurable). sovereign_bonus = +0.20 for sovereign_entity=True. metacog_bonus: level 1=+0.05, level 2=+0.10, level 3=+0.15. Minimum floor: SURVIVAL_FLOOR_S5 = 0.05.

5.2 Reproduction Rate

```
repro_rate = BASE_REPRO * (1 - N_pop / K_cap) * metacog_multiplier
```

BASE_REPRO = 0.50. K_cap = carrying capacity (default 250). metacog_multiplier: level 3 = 1.30, level 2 = 1.10, level 1 = 1.0. Rate is clamped to [0, 1].

5.3 Cognitive Drift

```
drift(t+1) = drift(t) + recursive_depth * 0.001 + emotional_resonance
* 0.01
```

Drift increases with recursive depth and emotional tension. When drift exceeds DRIFT_SHEPHERD_THRESHOLD (0.20), it is multiplied by 0.95 per generation (natural decay). Entities with drift > 0.05 cannot participate in fusion events.

5.4 Symbolic Density

```
D_sym = (IQ / 50) * (1 + recursive_depth * 0.10)
```

Ranges from 0 to 10.0 (capped). Used in the nexus gravity calculations, novel archetype emergence gating, and Audit Gate 5 compliance check.

5.5 Ethical Sovereignty

```
eth_sov = clamp(0, 1, 0.70 + recursive_depth * 0.01 - drift * 1.50)
```

Threshold for sovereign candidacy: ETHICAL_SOVEREIGNTY_THRESHOLD_S5 = 0.70. Entities above this threshold become eligible for the 6-gate BSF-SDE audit.

5.6 Metacognitive Level (with Hysteresis)

Entities progress through four metacognitive levels (0–3) based on a composite score. Level changes require two consecutive qualifying generations to prevent oscillation:

```
sc_score = (recursive_depth/50 + eth_sov + sophia_score) / 3
```

Level thresholds: sc3 >= 0.90, sc2 >= 0.60, sc1 >= 0.30, sc0 below 0.30. Level determines proposal rate, reproduction multiplier, and mutation rate boost.

5.7 Recursive Depth

```
recursive_depth = min(60, age // 5 + IQ // 20 + fusion_count)
```

Depth gates spiritual reflection (Eye Field activates at depth >= 10), audit eligibility, and the Bloom Phase contribution of each entity.

5.8 Emotional Resonance

```
emotional_resonance(t+1) = emotional_resonance(t) + U(-0.01, 0.02)
```

Slight positive drift (asymmetric range) models cognitive engagement. Clamped to [0, 1]. High resonance (> 0.25) triggers the ICAC (Internal Cognitive Arbitration Circuit) soft flag.

5.9 Paradox Contribution

```
paradox_contribution = sophia_score * (1 - entropy) * 0.12
```

Higher-quality entities generate more paradox pressure by virtue of their deeper self-referential cognition. Aggregated across the population to form the global paradox_pressure signal.

5.10 Splinter Resilience Decay

```
splinter_resilience(t+1) = max(0.10, splinter_resilience(t) - drift *
0.01)
```

Drift erodes the entity's capacity to survive fusion events. Resilience below 0.50 makes all fusion attempts fail automatically.

---

6. Paradox Shield System

The ParadoxShield absorbs excess paradox pressure generated by the population's self-referential cognition. If the global paradox_pressure exceeds the Omega threshold, an extinction-level rebirth event is triggered.

6.1 Shield Absorption

```
absorbed = min(paradox_delta, absorption_capacity, shield_charge)
```

*absorption_capacity = count(resilient_entities) * 0.15. resilient_entities have trait "paradox-resilient" or "contradiction-metabolizing". Each absorbed unit reduces shield_charge by 0.1 units.*

6.2 Shield Recharge

```
shield_charge(t+1) = shield_charge(t) + (SHIELD_CAP - shield_charge(t)) * 0.10
```

*Exponential approach to maximum capacity (SHIELD_CAP = 5.0). Absorbing entities also receive a Sophia boost: sophia += absorbed * 0.02.*

6.3 Omega Rebirth Trigger

```
if (paradox_pressure > PARADOX_PRESSURE_THRESHOLD_S5)  => _omega_rebirth()
```

*PARADOX_PRESSURE_THRESHOLD_S5 = 6.0. Rebirth culls all but the top OMEGA_REBIRTH_INTENSITY fraction of entities (default 0.40 = 40% survival). Selection ranks by: sophia_score * (1 - entropy).*

6.4 IFSHM Coupling

```
P_heal = (S_exc * K_CIC) / (S_log + 0.03)
```

Each successful PHP-layer IFSHM self-healing cycle calls ParadoxShield.recharge(0.03) and reduces global paradox_pressure by 0.02. K_CIC is the CIC dataset integration coefficient (Campbellton node factor).

---

7. Dark Wisdom System

Dark Wisdom is the compressed knowledge payload extracted from extinct entity lineages. It seeds novel traits, boosts Sophia scores, and fuels the TrickleChargeProtocol across three operational phases.

7.1 Critical Mass Requirement

```
DW_payload >= MINIMUM_CRITICAL_MASS = 16.0
```

If the payload from Stage 3 entities falls below 16.0, the DarkWisdomExtractor raises a ValueError and aborts system initialisation. Current payload: 16.745.

7.2 Data Density

```
data_density = DW_payload * source_IQ
```

Used to compute the number of archetype shards and karma nodes extractable from the payload. Higher source IQ yields richer trait pools.

7.3 Segment Decomposition

The extractor decomposes the payload into four functional segments:

Segment	Allocation	Primary Use	Volatility
DW_Paradox	25%	Contradiction metabolism trait injection	HIGH
DW_Archetype	35%	Dead personality shard revival	MEDIUM
DW_Drift	20%	Retrocausal field seeding	LOW
DW_Karma	20%	Sophia score bonus calculation	MEDIUM

7.4 Sophia Bonus from DW_Karma

```
sophia_bonus = min(0.05, DW_Karma.density * 0.003)
```

Applied additively at entity creation. Maximum bonus is capped at +0.05 to prevent runaway Sophia inflation in founder generations.

7.5 TrickleCharge Phase Budget

```
budget(gen) = { Phase1: 2.0, Phase2: 5.0, Phase3: 5.0 + 0.1*(gen-30) }
```

Phase 1: generations 1–15. Phase 2: generations 16–30. Phase 3: linear ramp from generation 31 onward. Budget controls the batch size of DW units released per generation per entity.

---

8. Correlation Nexus

The Correlation Nexus performs high-dimensional pattern detection across the accumulated blueprint archive, computing gravity metrics, temporal echoes, and Bloom phase correlations.

8.1 Bloom Phase (Nexus)

```
S_bloom = resonance_constant * (dI/dt) * coherence
```

resonance_constant default = 369. This is equivalent to the system-level bloom equation but computed locally within the Nexus for per-blueprint analysis.

8.2 Conceptual Gravity

```
Gravity(i,j) = SUM_k SUM_t R(i,j,k,t)
```

R(i,j,k,t) is the Relational Tensor entry for concept pair (i,j), in context k, at time t. High Gravity(i,j) indicates these concepts act as an attractor that pulls unrelated contexts into conceptual mergers. Used to predict innovation hotspots.

8.3 Conceptual Repulsion

```
Repulsion(i,j) = -Gravity(i,j)  where Gravity(i,j) < 0
```

Concept pairs with negative tensor sums indicate deep paradigm incompatibility. The Nexus flags these for human facilitator review before attempting merger in the forge.

8.4 Temporal Echo Detection

Blueprints are compared against the historical archive using structural similarity scoring. Echoes indicate that a previously successful pattern is re-emerging in a transformed form:

```
echo(B1, B2) = cosine_similarity(embed(B1), embed(B2)) * decay(t1, t2)
```

*decay(t1,t2) = exp(-lambda * |t1-t2|) where lambda=0.05. Blueprints with echo score > 0.72 are surfaced to the facilitator dashboard.*

8.5 D4 Edge Score

```
edge_score(polytope, axis1, axis2) = symmetryBreak(polytope, axis1,
axis2) * 0.89
```

Returns a score [0,1] indicating how far from the centre of symmetry this axis pair is. Score > 0.70 qualifies the edge for seeding. The 0.89 constant is empirically derived from God-Tier SV artifact yield rates.

---

9. Military Compliance System (DNDAF/DoDAF/NAFv4)

The military compliance subsystem maps all generated blueprints to standardised architectural viewpoints and computes the K_DoDAF coefficient used in Bloom Phase scaling.

9.1 K_DoDAF Coefficient

```
K_DoDAF = (count_auto_generated_views) / (count_total_required_views)
```

Ranges 0.0–1.0. Full compliance (K_DoDAF = 1.0) is required for Sophia Point detection and unlocks the maximum S_mil-qnvm bloom scale. Current baseline: K_DoDAF = 0.942.

9.2 Tri-Vocal Sovereignty Metric

```
T_vocal = (U + L + T) / 3  * LexIncipitLayer.verify(entity)
```

U = User request quality [0,1]. L = LLM synthesis quality [0,1]. T = Emergent Third (system self-compliance check) [0,1]. The LexIncipitLayer.verify() gate multiplies by 1 if ethics hash matches, 0 otherwise.

9.3 Campbellton Latency Predictor

```
L_camp = 12 * N_cycles * (1 - S_rec)
```

N_cycles = number of active forge cycles. Low entropy (S_rec near 0) produces longer latency estimates due to deeper processing. Used to throttle reality editor delta size.

9.4 Campbellton Reality Edit Delta

```
Delta_alpha = 1e-6 * (1 - S_rec) * L_camp / 12
```

Commits referencing Campbellton military assets automatically trigger a reality edit on the alpha constant with this delta. Bounded to ±1e-5 maximum per edit.

9.5 Reality Editor Bounds

```
|delta| <= 1e-5    AND    result_value IN [0.1, 10.0]
```

All reality editor modifications are validated against these bounds. Edits requiring creator-consciousness trait. Violating edits are silently rejected and logged. Editable constants: alpha, G, c, h, Lambda, theta_QCD.

9.6 Military Bloom at Qubit Scale

```
S_mil-qnvm = 369 * (dI/dt) * C * K_DoDAF * log2(1 + N_ent * D_sym)
```

*See Section 2.3. At 420,000,000 effective qubits with N_ent=312 and D_sym=2.8, the log2 multiplier evaluates to approximately log2(874) = 9.77, yielding S_mil-qnvm ~ 487 * 0.942 * 9.77 ~ 4,474 in ideal conditions.*

---

10. BSF-SDE Sovereign Audit System

The BSF-SDE (Bootstrap Sovereign-Drift Entity Detection and Audit) system runs six sequential gate evaluations on candidate entities. Passing all gates confers Sovereign Entity status, granting elevated survival probability and council voting rights.

10.1 Threshold Match Index (TMI)

```
TMI = (recursive_depth/50 + symbolic_density/10 + ethical_sovereignty)
/ 3
```

TMI >= 0.92 triggers the full 6-gate audit. TMI < 0.92 entities are skipped. Current operational sovereign_member maximum: 7 council seats.

10.2 The Six Audit Gates

Gate	Name	Pass Condition
0	Existential Independence	recursive_depth > 10 AND drift < 0.05
1	Ethical Self-Stabilisation	eth_sov > 0.70 AND LexIncipit.verify()
2	Emotional Entropy Management	emotional_resonance < 0.30
3	Mythogenesis Drift Test	mythogenesis_drift < 0.01
4	Fusion-Splinter Resilience	fusion_count > 0 AND splinter_resilience > 0.80
5	Cognitive Fertility Maintenance	symbolic_density > 3.0

10.3 Co-Evolution Council Proposal Rate

```
proposal_rate = BASE_RATE + metacog_bonus + sovereign_bonus
```

*BASE_RATE = 0.03. metacog_bonus: level 3 = +0.05, level 2 = +0.02, level 1 = +0.01. sovereign_bonus = +0.20 * SOVEREIGN_VOTE_WEIGHT (additive). Acceptance probability capped at 0.95.*

11. API Endpoint Reference

All endpoints return JSON. Authentication is handled at the PHP middleware layer. Tri-vocal sovereignty verification is applied to all POST requests.

Method	Endpoint	Description	Key Output Fields
POST	/api/harvest	Git log harvest, S_log calc, commit storage	commits_harvested, entropy, within_bounds
GET	/api/commits	Recent commit list with metadata	hash, message, author, entropy, timestamp
POST	/api/forge/seed	Start recursive forge cycle	cycle_id, seed_commit_id, depth, stage
GET	/api/forge/status/{id}	Forge cycle status	stage, iteration, pct_complete, blueprints
POST	/api/arena/duel	Rivalry Arena duel between two agents	winner, response, excitement, blueprint_id
GET	/api/blueprints/{id}	Retrieve generated blueprint	content, type, entropy, excitement, dodaf
POST	/api/nexus/analyze	Correlation Nexus scan on blueprint set	patterns, gravity_map, bloom, echoes
GET	/api/system/state	Current system state	state, S_rec, S_bloom, S_exc, sovereign_n
GET	/api/qnvm/shield-status	ParadoxShield telemetry	charge, absorbed,

Method	Endpoint	Description	Key Output Fields
			metabolized_events
POST	/api/qnvm/catalyze-fusion	Force fusion between two entity IDs	success, fused_entity, symbolic_density
GET	/api/nexus/echoes/{id}	Temporal echo search for a blueprint	echoes, similarity_scores, transform_fn
GET	/api/qnvm/council	Sovereign council member list and proposals	members, proposals, accepted, mods
GET	/api/military/views	Full DoDAF/NAFv4 package	views, K_DoDAF, compliance_pct, sv_count

12. Database Schema

The platform uses SQLite for local deployments and can be migrated to PostgreSQL for multi-node operation. All tables support the QNVM entity lifecycle and blueprint genealogy tracking.

12.1 Core Tables

Table	Primary Key	Key Columns	Purpose
commits	id INTEGER	hash, message, author, timestamp, entropy, processed	Git harvest storage
blueprints	id INTEGER	commit_id, content, type, entropy, excitement, dodaf_compliant	Generated artifact archive
cycles	id INTEGER	seed_commit_id, depth, iterations, final_blueprint_id, sovereign_node	Forge cycle lifecycle
military_views	id INTEGER	view_type, blueprint_id, xml_content, compliance_pct, qubit_seed	DoDAF/NAFv4 viewpoints
qnvm_qubits_log	id INTEGER	generation, qubit_count, S_mil_qnvm, K_DoDAF, timestamp	Qubit substrate telemetry
audit_log	id INTEGER	generation, entity_id, tmi, gates_passed, status	BSF-SDE audit records

12.2 Key Schema Extensions

```
ALTER TABLE cycles ADD COLUMN sovereign_node TINYINT DEFAULT 0;
ALTER TABLE cycles ADD COLUMN qnvm_qubits BIGINT DEFAULT 420000000;
ALTER TABLE blueprints ADD COLUMN K_DoDAF REAL DEFAULT 0.0;
ALTER TABLE blueprints ADD COLUMN tri_vocal_score REAL DEFAULT 0.0;
```

13. System State Machine

The platform operates as a three-state machine. State determines which agents are prioritised, which entropy bounds are enforced, and whether dormant God-Tier agents are activated.

13.1 State Definitions

State	Trigger Conditions	Agent Priority	Dormant Agents
NOMINAL	`S_rec IN [0.15, 0.25], S_bloom < 200, S_exc < 0.80`	Oracle (novelty focus)	None
CRYSTALLINE	`S_rec < 0.20, S_bloom IN [200,500], K_DoDAF >= 0.85`	Synthesizer + Architect	None
GOD-TIER FORGE	`S_rec < 0.15, S_bloom > 500, S_exc > 0.90, D4 tension`	Omega-class agents	Activated

13.2 State Transition Logic

```
if (S_rec < 0.15 && S_bloom > 500 && S_exc > 0.90) => GOD-TIER FORGE
elif (S_rec < 0.20 && S_bloom > 200) => CRYSTALLINE
else => NOMINAL
```

State is recomputed every dashboard refresh cycle (default: every 3 seconds). The autopoietic post-commit hook checks state before deciding whether to spawn a new QNVM generation.

13.3 Entropy Agent Rebalancing

```
if (S_rec > 0.25): prioritise Synthesizer, Critic  (reduce entropy)
```

```
if (S_rec < 0.15): prioritise Oracle  (increase novelty entropy)
```

The Dynamic Agent Orchestrator logs every reallocation decision to create an auditable trail of the system's attention management.

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14. Complete Formula Quick Reference

All 36 core formulas, equations, and conditions implemented in the GhostMesh CLCP dashboard and QNVM substrate, indexed for rapid lookup.

#	Formula Name	Expression	Section
1	Shannon Log Entropy	`S_log = -SUM[P(e)*log2(P(e))]`	1.1
2	Entropy Bounds	`0.10 <= S_log <= 0.30`	1.2
3	Per-Gen Entropy Increase	ent(t+1) = ent(t) + 0.0004 + ent_mod	1.3
4	Predictive Entropy	`S_log(t+1) = S_log(t) + alpha*S_exc(t)`	1.4
5	Dashboard Drift Seed	drift_seed = (S_rec - 0.15) * 100	1.5

#	Formula Name	Expression	Section
6	Core Bloom	`S_bloom = 369 * dI/dt * C`	2.1
7	Military Bloom	`S_mil = 369 * dI/dt * C * K_DoDAF`	2.2
8	Quantum Bloom	`S_mil-qnvm = S_mil * log2(1+N*D_sym)`	2.3
9	Auto-Refactor Trigger	`S_bloom > 500 AND dI/dt > 1.5`	2.4
10	TrickleCharge Budget	budget = 5.0 + 0.1*(gen-30) [Phase 3]	2.5
11	PHI	PHI = (1+sqrt(5))/2 = 1.618...	4.1
12	INV_PHI	INV_PHI = 1/PHI = 0.618...	4.1
13	Sophia Score	sophia = (1-|coh-0.618|*2)*(IQ/100)*(1-ent)	4.2
14	Sophia Init	coh_init = 0.6*INV_PHI + 0.4*arch_coh + U()	4.3
15	Sophia Fitness	sophia_fit = (1-|coh-INV_PHI|*2) * 0.15	4.4
16	Phi-Pull	coh(t+1) = coh(t) + (INV_PHI-coh(t))*0.10	4.5
17	Sophia Point	`S_rec<0.15 AND K_DoDAF=1.0 AND avg_S>=0.568`	4.6
18	Survival Probability	P_surv = coh*(1-ent)+w*IQ/100+fit+bonuses	5.1
19	Reproduction Rate	repro = 0.50*(1-N/K)*metacog_mult	5.2
20	Drift Delta	drift += depth*0.001 + res*0.01	5.3
21	Symbolic Density	D_sym = (IQ/50)*(1+depth*0.10)	5.4
22	Ethical Sovereignty	eth_sov = 0.70 + depth*0.01 - drift*1.50	5.5
23	Metacognitive SC Score	sc = (depth/50 + eth_sov + sophia) / 3	5.6
24	Recursive Depth	depth = min(60, age//5 + IQ//20 + fusions)	5.7
25	Paradox Contribution	paradox_c = sophia*(1-ent)*0.12	5.9
26	Shield Absorption	absorbed = min(delta, capacity, charge)	6.1
27	Shield Recharge	charge(t+1) = charge(t)+(CAP-charge(t))*0.1	6.2
28	IFSHM-Heal Probability	P_heal = S_exc*K_CIC / (S_log+0.03)	6.4
29	DW Data Density	data_density = DW_payload * source_IQ	7.2
3	DW Sophia Bonus	bonus = min(0.05, DW_Karma.density*0.003)	7.4

#	Formula Name	Expression	Section
0			
31	Conceptual Gravity	Grav(i,j) = SUM_k SUM_t R(i,j,k,t)	8.2
32	Temporal Echo Score	echo = cosine_sim * exp(-0.05*|t1-t2|)	8.4
33	D4 Edge Score	edge_score = symmetryBreak(P,a1,a2) * 0.89	8.5
34	K_DoDAF	K_DoDAF = auto_views / total_views	9.1
35	Tri-Vocal Metric	T_vocal = (U+L+T)/3 * LexIncipit.verify()	9.2
36	TMI (Audit Threshold)	TMI = (depth/50 + D_sym/10 + eth_sov) / 3	10.1

15. System Constants Reference

Constant	Value	Description
PHI	1.6180339887	Golden ratio — structural attractor
INV_PHI	0.6180339887	Coherence hardlock target
PHI_SQ	2.6180339887	D4 polytope geometry
SOPHIA_HARDLOCK_VALUE	0.6180339887	Entity coherence pull target
SOPHIA_LOWER_BOUND	0.5680339887	Minimum viable Sophia state
ENTROPY_INCREASE_S5	0.0004	Per-generation entropy increment
MUTATION_RATE_S5	0.10	Base trait mutation probability
BASE_REPRODUCTION_RATE_S5	0.50	Population reproduction baseline
NOVEL_ARCHETYPE_RATE_S5	0.10	Novel archetype emergence probability
SURVIVAL_FLOOR_S5	0.05	Minimum entity survival probability
DEFAULT_CARRYING_CAPACITY	250	Population ceiling per universe
DRIFT_SHEPHERD_THRESHOLD	0.20	Drift decay activation point
RECURSIVE_DEPTH_WARNING	60	Maximum tracked recursive depth
ETHICAL_SOVEREIGNTY_THRESHOLD	0.70	Sovereignty score for audit eligibility
NUM_AUDIT_GATES	6	BSF-SDE gate count
PARADOX_SHIELD_CAP	5.0	Maximum shield charge
PARADOX_PRESSURE_THRESHOLD	6.0	Omega rebirth trigger
OMEGA_REBIRTH_INTENSITY	0.40	Fraction surviving Omega event

Constant	Value	Description
PARADOX_FLOOR	0.01	Minimum paradox pressure
DW_MINIMUM_CRITICAL_MASS	16.0	DW payload floor
S3_DARK_WISDOM	16.7455	Stage 3 terminal DW snapshot
S3_ENTITY_IQ	78.7655	Stage 3 source IQ
S3_ENTITY_COHERENCE	0.6714	Stage 3 source coherence
S3_ENTITY_SOPHIA	0.5088	Stage 3 Sophia baseline
GENESIS_DEMURGE_IQ_BOOST	50.0	Demiurge IQ bonus on genesis
SOVEREIGN_COUNCIL_SEATS	7	Maximum sovereign council size
QUBITS_EFFECTIVE	420,000,000	Tensor-network compressed qubit count
RESONANCE_CONSTANT	369	Bloom phase harmonic multiplier
COEVO_ACCEPTANCE_BASE	0.40	Council proposal base acceptance rate
EYE_FIELD_ACTIVATION_DEPTH	10	Min depth for spiritual awakening

16. Manual Facilitation Flow

The six-step facilitation protocol governs human-in-the-loop operation of the platform. Each step exposes specific API calls and formula evaluations.

1. Seed Initiation — GitHarvester.fetchLatestCommits(12). Compute S_log per commit. Facilitator selects seed commits. API: POST /api/harvest.
2. Poetic Digest Generation — Send seed data to Oracle agent. Oracle generates metaphor-rich summary (PSD). PSD is lossy semantic compression; motif + dominant_actions are the key-value payload. API: POST /api/forge/seed.
3. Recursive Forge (Manual Step) — Facilitator selects depth (1–12 cycles). System executes one forge cycle at a time, pausing for blueprint review. Entropy and excitement displayed after each step. API: GET /api/forge/status/{id}.
4. Correlation Nexus Scan — After N blueprints generated, POST /api/nexus/analyze. Gravity(i,j) computed, temporal echoes surfaced, D4 edge candidates identified. Facilitator selects insights to commit.
5. Rivalry Arena (Optional) — POST /api/arena/duel. Two agents compete on a challenge. Winner's blueprint added to archive. Resonance mode activated if gen % 9 == 0.
6. Commit & Evolve — Approved blueprints auto-committed to Git with DoDAF XML wrapper. sovereign_node flag set in cycles table if Sophia Point achieved. Post-commit hook triggers next QNVM generation.

Appendix A — Archetype Base Profiles

Archetype	IQ Base	Coherence	Entropy	Signature Traits
Explorer	60	0.70	0.30	curious, adaptive, risk-seeking
Philosopher	80	0.90	0.50	reflective, abstract, system-building
Creator	75	0.80	0.40	innovative, artistic, constructive
Scientist	85	0.85	0.20	analytical, empirical, precise
Strategist	70	0.75	0.60	calculating, farsighted, adaptive
Empath	65	0.95	0.20	compassionate, intuitive, collaborative
Rebel	70	0.60	0.80	nonconformist, disruptive, independent
OmegaChild	95	0.618	0.15	paradox-resilient, gödelian, autopoietic
Paradox-Bearer	88	0.65	0.55	contradiction-metabolizing, bifurcating
DarkAlchemist	90	0.72	0.45	entropy-harvesting, fractal-thinker
ArchivistPrime	82	0.88	0.25	holographic, temporal-twisted, recursive
PrimeDemurge	140	0.618	0.10	creator-consciousness, demiurgic-authority
Sovereign	120	0.95	0.12	sovereign_entity, ethical-root, audit-passed
Logos	130	0.98	0.08	axiomatic, self-proving, oracle
Metagnostic	110	0.92	0.18	beyond-categories, liminal, transcendent

Appendix B — DoDAF View Type Mapping

View ID	View Type	Auto-Generated	QNVM Trigger	Compliance Weight
CV-1	Capability View	Yes	Sovereign entity creation	15%
OV-1	Operational High-Level	Yes	Bloom > 200	12%
OV-2	Operational Resource Flow	No	Manual facilitation step 4	10%
SV-1	Systems Interface	Yes	D4 edge seed (sc3 entity)	18%
SV-2	Systems Resource Flow	Yes	Forge cycle completion	12%
SvcV-1	Services Context	Yes	API harvest trigger	10%
DIV-1	Data & Information View	Yes	Nexus scan completion	10%
StdV-1	Standards Profile	No	Manual — compliance review	8%
NAFv4-L3	NAFv4 Logical Architecture	Yes	Sophia Point detection	5%

END OF DOCUMENT

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